

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : {FACULTYNAME}

Semester : SEMESTER - 6

Subject Code & Name: BTMC601 - MANUFACTURING PROCESSES-II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The process of improving the concentricity and correcting the shape of a grinding wheel is known as: (a) Dressing (b) Shaping (c) Truing (d) None of these
- 2 Aluminum oxide has ----- friability than/as silicon carbide. (a) more (b) less (c) same (d) None of these
- 3 When shear angle is more, the cutting power required will be -----.(a) more (b) less (c) unaffected (d) None of these
- 4 For the formation of non-homogeneous chips, following material properties are favorable: (a) Yield strength increases with temperature (b) Good thermal conductivity (c) Both 'a' and 'b' (d) None of these
- 5 It is essential that the instrument used for the measurement of cutting force should have the following characteristics: (a) high rigidity and less natural frequencies (b) Less rigidity and high natural frequencies cutting conditions (c) High rigidity and high natural frequencies (d) None of these
- 6 Crater wear typically occurs on which surface of the cutting tool? (a) Flank (b) Rake (c) Both 'a' and 'b' (d) None of these
- 7 ----- is the toughest cutting tool material. (a) Diamond (b) Ceramics (c) CBN (d) HSS
- 8 What is/are the purpose(s) of applying tool coatings in metal cutting?(a) To reduce friction between tool and chip (b) To decrease tool wear (c) Both 'a' and 'b' (d) None of these
- 9 ----- atomization usually results in more spherical particles.(a) Gas (b) Water (c) Both 'a' and 'b' (d) None of these
- 10 During powder compaction, the density within the part can vary considerably because of friction between the following: (a) the metal particles in the powder (b) the punch surfaces and die walls (c) Both 'a' and 'b' (d) None of these
- 11 Which of the following processes involves the continuous production of a plastic profile by forcing molten plastic through a die? (a) Injection Molding (b) Blow Molding (c) Extrusion (d) Compression Molding

- 12 Acrylic fibers are produced by the following method:(a) Melt spinning (b) Wet spinning(c) Gel spinning (d) Dry spinning

Q2. Solve the following.

- A) Define friability of a grinding wheel. Explain the factors affecting friability. Give the applications of various conventional and superabrasives. 6
- B) What is the importance of grinding wheel wear? Explain the mechanisms involved. Show the different wear regions for a grinding wheel graphically. 6

Q3. Solve the following.

- A) Show all the angles and nose radius of a single point cutting tool on the different views of a single point cutting tool. Explain the importance of auxiliary cutting edge angle, rake angle, clearance angle, and nose radius. 6
- B) Derive the mathematical expression for the shear angle in orthogonal cutting in terms of rake angle and cutting ratio. Also, derive the formula for undeformed chip thickness. 6

Q4. Solve Any Two of the following.

- A) What is significance of temperature generated during machining? What is the basic mechanism of temperature generation? Describe the principle of working of tool-work thermocouple with the help of a schematic diagram. 6
- B) Explain the basic requirements of cutting tool materials. What are the major classes of cutting tool materials? Arrange them in the increasing order of toughness. Define hot hardness and show the variation of hardness with temperature for different tool materials. 6
- C) If in turning of a steel rod by a given cutting tool, the tool life decreases from 80 min to 20 min due to increase in cutting velocity from 60 m/min to 120 m/min, then at what cutting velocity the life of that tool under the same condition and environment will be 40 min? 6

Q5. Solve Any Two of the following.

- A) Define atomization method of producing metal powders. Explain gas atomization with the help of a schematic diagram. Compare gas atomization with water atomization. 6
- B) What is isostatic pressing? Explain hot isostatic pressing with the help of a suitable diagram. Compare hot isostatic pressing with cold isostatic pressing. 6
- C) What is sintering? What are the process variables? Explain the sintering mechanisms with the help of schematic diagrams. 6

Q6. Solve Any Two of the following.

- A) Classify plastic-shaping processes according to the resulting product geometry. Discuss extrusion process w.r.t. basic principle of working, equipment details, and applications. 6
- B) Explain the injection molding process with a neat and labelled diagram. Also, give the applications of injection molding. What is the difference between injection molding and reaction injection molding? 6
- C) Discuss transfer molding and compression molding processes with the help of schematic diagrams. Give the applications of both. 6